Enthusiast COURSE

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL : 2022

Test Type :**PRACTICE TEST** Test Pattern :**NEET (UG)**

TEST DATE : ________

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	1	1	2	2	4	4	2	2	2	1	1	3	4	2	1	4	2	4	1	3	2	4	2	2	3	2	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	2	4	1	3	4	3	4	4	3	3	2	3	1	1	3															

HINT – SHEET

1. Ans (2)

$$R=\frac{\sqrt{2mK}}{qB}\Rightarrow R_B=R_P\Rightarrow\frac{2m_DK_D}{q_D^2B^2}=\frac{2m_PK_P}{q_P^2B^2}$$

$$KP=\frac{m_D}{m_P}\times K_D=\frac{2m_P}{m_P}\times 1=2\text{ MeV}$$

2. Ans (3)

$$B=\frac{\mu_0 I}{2r}=I=\frac{q}{T}=q.\frac{\omega}{2\pi}\Rightarrow B=\frac{\mu_0 q\omega}{4r\pi}$$

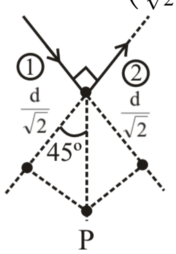
3. Ans (1)

$$|\vec{M}|=\sqrt{M_1^2+M_2^2+2M_1M_2\cos\theta}$$

$$=2M\cos\frac{\theta}{2}\ \text{If }\theta\uparrow M'\downarrow$$

4. Ans (1)

$$B_1=\frac{\mu_0 I}{4\pi\left(\frac{d}{\sqrt{2}}\right)}(\cos 0^\circ-\cos 45^\circ)\otimes$$



$$B_1=B_2=\frac{\mu_0 I}{4\pi\left(\frac{d}{\sqrt{2}}\right)}(\cos 0^\circ-\cos 45^\circ)\otimes$$

$$B=2B_1=2B_2=\frac{\mu_0 I}{\sqrt{2}\pi d}\left(1-\frac{1}{\sqrt{2}}\right)\otimes$$

5. Ans (2)

$\vec{\tau}=\vec{M}\times\vec{B}$ for equilibrium $\tau=0$

$\Rightarrow$ $\theta=0^\circ$ stable equilibrium

$\Rightarrow$ $\theta=180^\circ$ unstable equilibrium.

6. Ans (2)

in magnetic field $qvB=\frac{mv^2}{r}\rightarrow V=\frac{qBr}{m}$

when electric field is switched on $qE=qvB$

$$E=\frac{qB^2r}{m}=10V/m$$

7. Ans (4)

$$K=\frac{q^2B^2R^2}{2m}$$

$$\Rightarrow\frac{K_\alpha}{K_\beta}=\frac{q_\alpha^2B^2R_\alpha^2}{2m_\alpha}\times\frac{2m_p}{q_p^2B^2R_p^2}$$

$$\frac{(2q_p)^2\ (2R)^2}{2\times 4m_p}\times\frac{2m_p}{q_p^2R^2}=4$$

$\therefore K_a=4K_p=4\times 1=4$ MeV

8. Ans (4)

$$B_1=\frac{\mu_0(2I)}{2R}\quad B_2=\frac{\mu_0(3I)}{2R}\qquad\therefore\vec{B}_1\perp\vec{B}_2$$

$$\therefore B_R=\sqrt{B_1^2+B_2^2}=\frac{\mu_0 I}{2R}\sqrt{2^2+3^2}=\frac{\mu_0 I\sqrt{13}}{2R}$$

9. Ans (2)

$$v_{max} = \frac{2\pi R}{T} = 2\pi nR$$

$$K_{max} = \frac{1}{2} m v_{max}^2$$

$$\Rightarrow K_{max} = \frac{1}{2} m 4\pi^2 n^2 R^2 = 2\pi^2 n^2 mR^2$$

10. Ans (2)

In stable equilibrium $U_i = -MB$

in unstable equilirbium $U_f = MB$

$\Delta U = U_f - U_i = 2MB \Rightarrow \Delta U = 0.128J$

11. Ans (2)

$$\vec{F}_{net} = \vec{0} \quad \vec{F}_{PQ} + \vec{F}_{QR} + \vec{F}_{RP} = \vec{0}$$

$$0 + \vec{F} + \vec{F}_{PR} = \vec{0}$$

$$\Rightarrow \vec{F}_{PR} = -\vec{F}$$

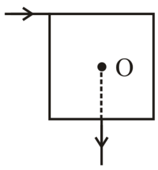
13. Ans (1)

If $B_{ext} = 0$ then for paramagentic materical

$M = 0$

14. Ans (3)

Magnetic field will be at centre due to wire.



15. Ans (4)

$\tau = BINA \sin\theta \quad \theta = 30^{\circ}$

$$B = \frac{\tau}{N\,I\,A \sin 30} = \frac{1.2 \times 10^{-2}}{1000 \times 1 \times 3 \times 10^{-4} \times 0.5} = 8 \times 10^{-2} T$$

16. Ans (2)

When field is parallel to the direction of motion of charge then magnetic force $f = qvB \sin 0° = 0$

17. Ans (1)

The ratio $\frac{M}{L}$ of is always $\frac{q}{2m}$

18. Ans (4)

$$T = 2\pi\sqrt{\frac{I}{MB_H}} \Rightarrow T \propto \frac{1}{\sqrt{B_H}}$$

$$\Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{B_H}{B_H - B}}$$

$$\Rightarrow \frac{6}{2} = \sqrt{\frac{B_H}{B_H - B}} \Rightarrow 9 = \frac{B_H}{B_H - B}$$

$$\Rightarrow \boxed{B = 32\mu T}$$

19. Ans (2)

$$\vec{F} = q\left(\vec{V} \times \vec{B}\right) \quad q = -3 \times 10^{-6}\,C$$

$$= -3 \times 10^{-6} \times \left[\left(3\hat{j} + \hat{k}\right) \times \left(-2\hat{j}\right)\right] \times 10^{6}$$

$$= 6\ (\hat{k} \times \hat{j})\ 6\ \left(-\hat{i}\right) = 6N \text{ in -x dir.}$$

20. Ans (4)

$B = \mu_0 nI$

$\mu_0 H = \mu_0 nI$

$H = nI = 10^4 \times 4 = 4 \times 10^4$ A/m.

23. Ans (2)

$$T = 2\pi\sqrt{\frac{I}{MB_H}} \qquad M' = 2M \qquad m' = \frac{m}{2}$$

$$I' = \frac{I}{2}$$

$$T' = 2\pi\sqrt{\frac{I}{2 \times 2MBH_4}} \quad , \quad T' = \frac{T}{2}$$

24. Ans (4)

Mass of positron (e^+) is very less.

25. Ans (2)

$$\frac{1}{2} mv^2 = qV \Rightarrow mv = \sqrt{2mqV}$$

$$R = \frac{mv}{qB} = \frac{\sqrt{2mqV}}{qB} = \frac{1}{B}\sqrt{\frac{2mV}{q}}$$

26. Ans (2)

$$r = \frac{mv}{qB} \propto v \quad V \uparrow r \uparrow$$

27. Ans (3)

$$R \propto \frac{\sqrt{T}}{B} \Rightarrow$$

$$\frac{R'}{R} = \sqrt{\frac{T'}{T} \cdot \frac{B}{B'}} = \sqrt{\frac{2T}{T} \cdot \frac{B}{3B}} = \sqrt{\frac{2}{9}}$$

$$\Rightarrow R' = \sqrt{\frac{2}{9}} R$$

28. Ans (2)

By the law find direction of force on current carring wire in magnetic field.

29. Ans (3)

$$2\pi r = \ell \Rightarrow r = \frac{\ell}{2\pi}$$

$$\tau = IAB = I \times \pi r^2 \times B = \pi IB \times \frac{\ell^2}{4\pi^2}$$

$$\Rightarrow I = \frac{4\pi\tau}{\ell^2 B}$$

30. **Ans (2)**

$W = MB(\cos\theta_1 - \cos\theta_2)$

$= MB(\cos 90^\circ - \cos 120^\circ)$

$= 3 \times 10^4 \times 4 \times 10^{-4}\left(0 + \frac{1}{2}\right) = 6J$

31. **Ans (2)**

$F_{PQ} = \frac{\mu_0 I\, I_0}{2\pi a}(a)$ (Repulsive)

$F_{RS} = \frac{\mu_0 I\, I_0}{4\pi\, a}(a)$ (Attractive)

$F_{Net} = F_{PQ} - F_{RS} = \frac{\mu_0 I\, I_0}{4\pi}$ Repulsive

32. **Ans (4)**

$\ell = 2\pi R = 4 \times 2\pi r \Rightarrow r = \frac{R}{4}$

For one turn $B = \frac{\mu_0 I}{2R}$

for 4 turns $B' = \frac{4\mu_0 I}{2r} = 16\frac{\mu_0 I}{2R}$ $B' = 16B$

33. **Ans (1)**

$\vec{B} = \frac{\mu_0 I}{2R}(\pm\hat{i}) + \frac{\mu_0 I}{2R}(\pm\hat{j}) + \frac{\mu_0 I}{2R}(\pm\hat{k})$

$|\vec{B}| = \sqrt{3}\frac{\mu_0 I}{2R}$

34. **Ans (3)**

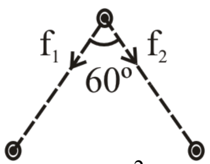
$W = MB[\cos 0^\circ - \cos 180^\circ] = 2MB = 2NIAB$

$W = 2 \times 200 \times 2 \times (3 \times 1.5 \times 10^{-4})$

$= 14.4 \times 10^{-3}$ J

$= 14.4$ mJ

35. **Ans (4)**



$f_1 = \frac{\mu_0 I^2}{2\pi d} = f_2$

$F = \sqrt{f_1^2 + f_2^2 + 2f_1 f_2\cos\theta} = \sqrt{3}f_1$

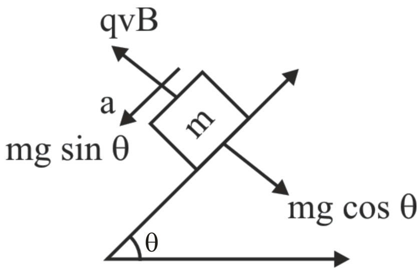
$F = \frac{\sqrt{3}\mu_0 I^2}{2\pi d}$

36. **Ans (3)**

$a = g\sin\theta$

Let velocity of block when

it leave contact with surface is v then



$q\,vB = mg\cos\theta \Rightarrow v = \frac{mg\cos\theta}{qB}$

By $v = u + at \Rightarrow \frac{mg\cos\theta}{qB} = 0 + g\sin\theta.\,t$

$t = \frac{1}{g\sin\theta} \times \frac{mg\cos\theta}{qB} = \frac{m\cot\theta}{qB}$

37. **Ans (4)**

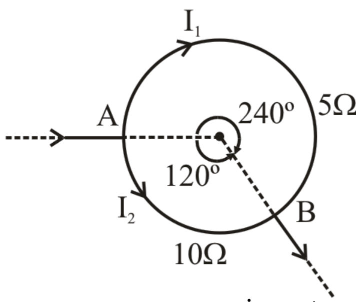
$R_1 < R_2 \Rightarrow \frac{m_1 v}{q_1 B} < \frac{m_2 B}{q_2 B} \Rightarrow \boxed{\frac{m_1}{q_1} < \frac{m_2}{q_2}}$

38. **Ans (4)**

$B = \frac{\mu_0 i}{4\pi d}(\sin\phi_1 + \sin\phi_2)$

$\frac{\mu_0 i}{4\pi d}(\sin 60 + \sin 45) = \frac{\mu_0 i}{8\pi d}(\sqrt{3} + \sqrt{2})$

39. **Ans (3)**



$I_1 = 10$ A $B_1 = \frac{\mu_0 i}{4\pi r} \times \frac{4\pi}{3} = \frac{\mu_0 10}{3r}$ $\otimes$

$I_2 = 5$ A $B_2 = \frac{\mu_0 i}{4\pi r} \times \frac{2\pi}{3} = \frac{\mu_0 5}{6r}$ $\odot$

$B_{Net} = \frac{\mu_0 10}{3r} - \frac{\mu_0 5}{6r} = \frac{5\mu_0}{2r}$ $\otimes$

41. **Ans (2)**

$B_1 = \frac{\mu_0 I_1 \alpha_1}{4\pi R}$ $B_2 = \frac{\mu_0 I_2 \alpha_2}{4\pi R}$

$\therefore I_1\alpha_1 = I_2\alpha_2$

So $\frac{B_1}{B_2} = 1$

42. Ans (3)

$$B_{Net} = \frac{\mu_0 i}{4\pi r} \times \frac{\pi}{6} - \frac{\mu_0 i}{4\pi \times 2r}\frac{\pi}{6} + \frac{\mu_0 i}{4\pi \times 4r} \times \frac{\pi}{6} - \ldots$$

$$= \frac{\mu_0 i}{24r}\left[1 - \frac{1}{2} + \frac{1}{4} \ldots\ldots\ldots \infty\right]$$

$$= \frac{\mu_0 i}{24r} \times \frac{2}{3} \qquad \left[S_\infty = \frac{a}{1-r} = \frac{2}{3}\right]$$

$$B_{Net} = \frac{\mu_0 i}{36r}$$

43. Ans (1)

$$R_{abc} = \frac{\rho\, 3\ell}{A}, \qquad R_{abc} = \frac{\rho \ell}{(A/3)} = \frac{3\rho\ell}{A}$$

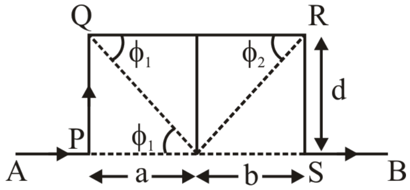
$$\Rightarrow I_{abc} = I_{adc} = i = \frac{I}{2}$$

$$B_{abc} = \frac{\mu_0 i}{4\pi R} \times \frac{3\pi}{2} = \frac{3\mu_0 i}{8R} = \frac{3\mu_0 I}{16R} \otimes$$

$$B_{abc} = \frac{\mu_0 i}{4\pi R}\frac{\pi}{2} = \frac{\mu_0 i}{8R} = \frac{\mu_0 I}{16R} \odot$$

$$B_0 = \frac{\mu_0 I}{16R}(3-1) \otimes \quad = \frac{\mu_0 I}{8R} \otimes$$

44. Ans (1)



$B_{AP} = B_{SB} = 0$

$$B_{PQ} = \frac{\mu_0 I}{4\pi d}[\sin\phi_1 + 0]$$

$$= \frac{\mu_0 I}{4\pi\left(\frac{d}{\tan\phi_1}\right)}\sin\phi_1 \quad = \frac{\mu_0 I}{4\pi d}\frac{\sin^2\phi_1}{\cos\phi_1}$$

$$B_{RS} = \frac{\mu_0 I}{4\pi d}\frac{\sin^2\phi_2}{\cos\phi_2}$$

$$B_{QR} = \frac{\mu_0 I}{4\pi d}[\cos\phi_1 + \cos\phi_2]$$

$$B_0 = B_{AP} + B_{PQ} + B_{QR} + B_{RS} + B_{QR}$$

$$= \frac{\mu_0 I}{4\pi d}\left[\frac{\sin^2\phi_1}{\cos\phi_1} + \cos\phi_1 + \frac{\sin^2\phi_2}{\cos\phi_2} + \cos\phi_2\right]$$

$$\frac{\mu_0 I}{4\pi d}(\sec\phi_1 + \sec\phi_2)$$

45. Ans (3)

$\because\ U = -MB\cos\theta$

$U_1 = -MB\cos 90^\circ = 0$

$U_2 = -MB\cos\theta$

$\Rightarrow\quad 0 < \theta < 90^\circ$

$-MB < U_2 < 0$

$U_3 = -MB\cos\theta = -MB$

$U_4 = -MB\cos\theta$, $90^\circ < \theta < 180^\circ$

$0 < U_4 < MB$

$\therefore\quad U_3 < U_2 < U_1 < U_4$